

Roshan Raj

Curriculum Vitae

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Research Interests

Interested in exploring the conceptual and mathematical foundations of quantum information science, particularly through quantum optics and photonic quantum technologies, with current interests in quantum algorithms, quantum error correction, quantum communication, and quantum metrology.

Education

M.Sc. Physics

Oct 2025 - Present

Universität Hamburg (UHH), Hamburg, Germany

Integrated M.Sc. Physics (5-year)

2019 - 2024

National Institute of Technology (NIT), Surat, India CGPA: 9.37 / 10.0

Thesis: *Quantum and Classical Applications of the Relativistic Relationship of Kinetic Energy and Momentum*. Guide: Dr. Vikash K. Ojha.

Briefly enrolled: PhD Physics, IISER Mohali (2025); B.Sc. Physics, Patliputra University, Patna (2018).

All India Higher Secondary (CBSE) Loyola High School, Patna

2018 | 77.2%

All India Secondary (ICSE) St. Paul's High School, Patna

2016 | 91.6%

Research Experience

Lie-Theoretic Investigation of Interferometer

2026 - Present

Supervisor: Dr. Schlawin, Institute for Quantum Physics (IQP), UHH, Hamburg

- A student project on Lie-theoretic interferometry

Integrated M.Sc. Thesis

2023 - 2024

NIT Surat & Supervisor: Dr. Ojha

- Tackled the relativistic anharmonic quantum harmonic oscillator and Maxwell-Jüttner distribution under a novel relativistic formalism.

Projects & Preprints

- Reviewing observables in classical and quantum mechanics. [osf.io/p2ufx](https://arxiv.org/abs/2305.12345) (2023).
- An open-ended story on quantization. [osf.io/92bvp](https://arxiv.org/abs/2305.12345) (2023).
- On Concepts of observable and indeterminacy in the quantum realm (2022)
- On the investigation of two non-neutral static bodies. [arXiv.2209.10641](https://arxiv.org/abs/2209.10641) (2022).
- The pictures of quantum dynamics. Unpublished [Book chapter](#) (Hindi) (2022).
- Dynamics of two objects considering the minimum total potential energy principle. [osf.io/ukx9t](https://arxiv.org/abs/2105.12345) (2021).

Selected coursework

At UHH: Quantum Computing and Information I & II (Prof. Dieter Jaksch) · Quantum Technology I & II (Prof. Ludwig Mathey) · Quantum Metrology and Sensing (Dr. Frank Schlawin) · Light fields & Quantum Correlations (Prof. Roman Schnabel) · Quantum Statistics (Prof. Michael Thorwart)
At IISER: General Relativity and Cosmology (Prof. Kinjalk Lochan)
At NIT: Quantum Field Theory, Mathematical Methods in Physics (Dr. Vikash K. Ojha) · Particle Physics (Prof. Ajay K. Rai)

Schools, Seminars & Workshops

- European Quantum Technology Summer School, AUAS Amsterdam (2026)
- ICTP PWF Bangladesh: School on Quantum Field Theory, ICTP Italy (Online, 2024).
- Talk: ‘On Unification in Mathematics,’ InterAct Seminar Series, SVNIT Surat (2021).
- Quantum Fields, Geometry & Representation Theory, ICTS-TIFR Bangalore (Online, 2021).
- Short Course in Math History - Prof. A. Viramani, IIT Gandhinagar (2021).
- Online: Special Relativity, Quantum Mechanics & Exp. Techniques -CCE IIT Kanpur (2020).
- National Workshops on MATLAB and Data Analysis, BBD NITM Lucknow (2020).

Achievements & Qualifications

- Joint CSIR-UGC National Eligibility Test (NET), Physical Sciences - Qualified (2025)
- GATE Physics - Qualified: 2023, 2024, 2025
- National Talent Search Examination (NTSE) - National Merit Scholarship, New Delhi (2016-24).
- Winner (Physics), InQuest 4.0 Researchathon, SCOSH, NIT Surat (2020).

Responsibilities & Positions

- Appointed Joint Academic Affairs Secretary, Academic Affairs Council, NIT Surat (2022).
- Student Coordinator, Quanta Seminar Series, Department of Physics, NIT Surat (2022).
- Co-Convenor, Society for Cultivation of Science and Humanities (SCOSH), NIT Surat (2021).
- Junior Graphic Designer, SCOSH Student Chapter, NIT Surat (2020).
- Student Volunteer, Unnat Bharat Abhiyan (UBA), Surat (2020).

Skills

Physics: Quantum Mechanics · Quantum Computing (+ Qiskit (basic)) · Mathematical Physics

Technical: Python (intermediate) · MATLAB · $\text{\LaTeX}$

Languages: Hindi (native) · English (proficient) · German (very beginner)

Other

- Self-published poetry collections: *KaalSangharsh* (2021), *Mano Vighn* (2023).
- Notes: ‘A Note on Consciousness’ (2026) · ‘On Einstein’s Theory of Gravitation’ (2025) · ‘On One of the Applications of Electrodynamics and Crystals’ (2025).

Declaration: *I hereby declare that the information given above is true to the best of my knowledge.*

Roshan Raj

(Roshan Raj)

Date: 05/06/2026.